

[Article preview](#)[Cited by \(442\)](#)[Metrics](#)[Expert Systems with Applications](#)**Date:** 1 May 2021[Show more](#) ▼Research article [Get rights and content](#) ↗

Multi-hour and multi-site air quality index forecasting in Beijing using CNN, LSTM, CNN-LSTM, and spatiotemporal clustering ☆

Rui Yan ^{a b} ✉, Jiaqiang Liao ^{a c} ✉, Jie Yang ^{a d} ✉, Wei Sun ^{a e f} ✉, Mingyue Nong ^a ✉, Feipeng Li ^a ✉[Show more](#) ▼ [Cite](#) [Add to Mendeley](#) [Share](#)[10.1016/j.eswa.2020.114513](#) ↗Access through **your organi...**[Purchase PDF](#)

Highlights:

- The AQI forecasting model uses deep learning and spatiotemporal clustering.
- The multiple-site forecasting models were developed for the next 1–6h.
- The overall forecasting for all the stations in Beijing through LSTM is optimal.
- Seasonal or spatial clustering-based forecasting is suitable for a season or cluster.
- CNN-LSTM and LSTM generally outperform CNN and BPNN.

Article preview

Abstract

[Introduction](#)[Section snippets](#)[References \(66\)](#)

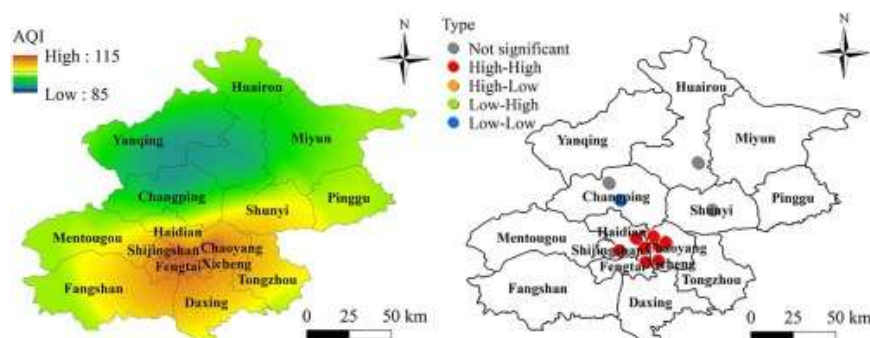
Abstract

Effective air quality forecasting models are helpful for timely prevention and control of air pollution. However, the spatiotemporal distribution characteristics of air quality have not been fully considered in previous model development. This study attempts to establish a multi-time, multi-site forecasting model of Beijing's air quality by using deep learning network models based on spatiotemporal clustering and to compare them with a back-propagation neural network (BPNN). For the overall forecasting, the performances in next-hour forecasting were ranked in ascending order of the BPNN, the convolutional neural network (CNN), the long

[Article preview](#)[Cited by \(442\)](#)[Metrics](#)

improved compared to the overall forecasting. For the spatial clustering-based forecasting, cluster 2 forecasting generally outperforms cluster 1 and the overall forecasting. Overall, either the seasonal or the spatial clustering-based forecasting is more suitable for the improvement of the forecasting in a certain season or cluster. In terms of model type, both the CNN-LSTM and the LSTM generally have better performance than the CNN and the BPNN.

Graphical abstract



[Download: Download high-res image \(202KB\)](#)

[Download: Download full-size image](#)

Keywords

LSTM; CNN; Forecasting; AQI; Spatiotemporal clustering

[< Previous article](#)

[Next article >](#)

Introduction

With urbanization and industrialization, air pollution in many countries and cities is increasingly serious. Previous studies have shown that long-term inhalation of air pollutants causes adverse effects on human health. Various air pollutants stimulate respiratory mucosa and induce respiratory diseases (Brunekreef & Holgate, 2002). Long-term exposure to atmospheric particles may cause impaired vascular endothelial functions and a series of cardiovascular diseases (N. Bai, Khazaei, van Eeden, & Laher, 2007). Atmospheric fine particles also affect the nervous system, increasing the risk of neurological diseases, such as Alzheimer's and Parkinson's diseases (Calderón-Garcidueñas, et al., 2008). It is estimated that the death toll due to outdoor and indoor air pollution in China is as high as 2.5 million

Forecasting models are such an air quality management tool, classified into numerical and statistical categories. Numerical models are based on atmospheric physical and chemical processes and use meteorological principles and mathematical methods to simulate the air quality, on a large scale, in both horizontal and vertical dimensions (An, et al., 2001). Since the middle of the 20th century, numerical models have been developed to a third generation, including Community Multiscale Air Quality Modeling System (CMAQ) (Streets, et al., 2007), Weather Research and Forecasting Model with Chemistry (WRF-Chem) (Tie, et al., 2007), and Nested Air Quality Prediction Modeling System (NAQPMS) (Z. Wang et al., 2014). These models depend on detailed data sets of pollutant emissions, require relatively complex calculations, and have high uncertainties in the forecasting results.

Statistical models do not consider atmospheric processes explicitly. Statistical models are used to develop air quality forecasting models from a data-driven perspective. The methods, such as autoregressive integrated moving average (ARIMA) (U. Kumar & Jain, 2010) and multiple linear regression (MLR) (Vlachogianni, Kassomenos, Karppinen, Karakitsios, & Kukkonen, 2011), have been widely used in short-term forecasting. Since Boznar, Lesjak, and Mlakar (1993) first established an SO₂ concentration forecasting model based on the Multi-Layer Perceptron (MLP) in 1993, machine learning has been widely applied, such as the artificial neural network (ANN) (Esen, Inalli, Sengur, & Esen, 2008c), the random forest (RF) (Rubal & Kumar, 2018), the classification and regression tree (CART) (Gocheva-Ilieva, Voynikova, Stoimenova, Ivanov, & Iliev, 2019), and the support vector regression (SVR) (Leong, Kelani, & Ahmad, 2019). The back-propagation neural network (BPNN) was used to predict concentrations of the pollutants and outperformed an MLR (Cai, Yin, & Xie, 2009). The wavelet neural network (WNN) combines the strengths of discrete wavelet transform and neural network processing and has been successfully applied to forecasting (Y. Bai et al., 2016, Esen et al., 2009). The adaptive neuro-fuzzy inference system (ANFIS) outperforms the ANN since the ANFIS inherits the interpretability of the fuzzy inference system and the learning ability of adaptive networks (Esen et al., 2017, Esen et al., 2008a, Prasad et al., 2016).

Machine learning is also combined with other methods to further improve forecasting performance. For example, a statistical weighted pre-processing method was developed to optimize ANN to use the

Article preview

Cited by (442)

Metrics

performance (S. Zhu, et al., 2017). Evolutionary neuro-fuzzy inference systems, such as the ANFIS with particle swarm optimization (ANFIS-PSO) and genetic algorithm (ANFIS-GA), produce better estimates compared with the classical ANFIS (Alizamir, Kisi, Muhammad Adnan, & Kuriqi, 2020). The gravitational search algorithm (GSA) can efficiently search for the optimal values of the unknown parameters of the least-square support vector regression (LSSVR). The integrated LSSVR-GSA enhances LSSVR's ability to estimate and forecast (Muhammad Adnan, et al., 2020). However, these models have drawbacks in feature extraction and generalization ability. Especially when the problem requires accurate prediction with big data, the forecasting ability of traditional statistical models is limited.

Deep learning is a powerful machine learning method suitable for big data, which has been widely applied in image understanding, speech understanding, natural language processing (Hao, Zhang, & Ma, 2016), and recently air quality forecasting. For example, an air quality prediction model based on long short-term memory (LSTM) appeared to have better performance than other traditional models such as ARIMA and SVR (X. Li, et al., 2017). The LSTM was combined with both the secondary decomposition and the optimization algorithm to establish a better hybrid model for AQI prediction in Beijing and Guilin (Q. Wu & Lin, 2019). The convolutional neural network (CNN), with successful applications in image analysis (Dong, Loy, He, & Tang, 2016), can account for the spatial distribution characteristics of air pollution. The CNN has been used to process the historical data of multiple sites in the form of a "class diagram" according to spatial locations and obtained higher prediction accuracy than the factorization-machine-supported neural network (FNN) (C. Wu, Li, Hou, Karimian, & Chen, 2018).

Although deep learning is promising to improve the forecasting accuracy of air quality, the accuracy of a single network may be limited when more complex problems are encountered. Many researchers have examined the combination of multiple deep learning networks (Karim et al., 2018, Quang and Xie, 2016). The well-received CNN-LSTM combines the advantages of both the CNN and the LSTM. The CNN can effectively extract features of grid data and the LSTM has excellent processing ability for time series data. For example, a deep CNN-LSTM model for PM_{2.5} forecasting in smart cities used the LSTM to analyze the features extracted by the CNN (C.-J. Huang & Kuo, 2018). A hybrid model based on the CNN-LSTM for ozone concentration prediction significantly reduced the error compared with the MLP and LSTM (Pak, Kim, Ryu, Sok, & Pak, 2018). However,

Cluster analysis is a common method to divide disordered data into several categories with high similarity to analyze the internal patterns (K. Wang, Qi, Liu, & Song, 2018). It can provide training samples with high similarity for a model, which not only reduces the training time but also enhances the generalization ability (D. Li, Jiang, & Cao, 2006). Cluster analysis has been used in air quality forecasting by selecting or dividing the input variables to enhance the data regularity (S. Chen et al., 2019, Liu et al., 2020). However, the spatiotemporal distribution may affect the air quality forecasting. Reports on spatiotemporal clustering-based air quality forecasting are limited. A study on PM_{2.5} forecasting of seven urban agglomerations in China based on the RNN-CNN ensemble model shows that the forecasting accuracy for different regions is diverse (J. Huang, Zhang, Du, Liu, & Cao, 2019). The seasonal nonlinear grey Bernoulli model outperformed the traditional nonlinear grey Bernoulli model in improving the prediction accuracy (Zhou, Wu, Ding, & Cheng, 2020). Therefore, air quality forecasting based on deep learning combined with spatiotemporal clustering is promising.

The objective of this study is to establish multi-time multi-site deep learning models (LSTM, CNN, CNN-LSTM) in hourly forecasting of air quality (AQI) in Beijing, China. It entails (1) spatiotemporal distribution characteristics analysis of AQI, with explanations of the cause and effect relations for air quality; (2) development and comparison of the overall forecasting based on LSTM, CNN, CNN-LSTM models; (3) combination of CNN, LSTM, CNN-LSTM models with spatiotemporal clustering and comparison with the overall forecasting; and (4) application of the proposed models to multi-site hourly AQI forecasting in Beijing. In this study, the spatiotemporal distribution characteristics of AQI and their impact on air quality forecasting are analyzed by coupling spatial or temporal cluster analysis with deep learning.

Organizational access

Get full-text access by signing in with your organisation

Access through your organization

Other access options

Section snippets

Air quality in Beijing

Beijing was selected as the study area because it has some of the most serious air pollution in the world. It is also the economic, social, and cultural center of China. It is in the north of North China Plain with Yanshan Mountains in the north and Taihang Mountains in the west. Its terrain is high in the northwest and flat in the East. It has shorter springs and autumns, hot and rainy summers, and cold and dry winters, because of the sub-humid warm temperate continental monsoon climate. Owing ...

Overall framework

In this study, three types of deep learning neural network models (CNN, LSTM, and CNN-LSTM) based on spatiotemporal clustering were developed for the Beijing AQI forecasting for 6h. After the spatiotemporal analysis of AQI in Beijing was conducted, all data were divided into several clusters in the temporal or spatial dimension. The AQI forecasting models were established based on either the total data or those in different spatiotemporal clusters. Different models were evaluated and compared ...

Spatiotemporal distribution characteristics of AQI

In the temporal dimension, the annual seasonal and monthly changes of the AQI were analyzed (Fig. 4a). The average AQI of Beijing from 2015 to 2016 was 105.42. There were obvious seasonal differences ($P < 0.001$) of the AQI in the two years, decreasing in the order of winter, spring, autumn, and summer. The summer average AQI (84.12 ± 52.37) was lower than the overall average. But the winter average AQI (125.05 ± 117.64) was significantly higher than the overall average. The average AQI of each ...

Discussion

In this study, the AQI forecasting models based on three deep learning methods, one shallow learning method, and spatiotemporal clustering were developed and compared. The results show that the LSTM and the CNN-LSTM generally perform better in multi-hour forecasting, whereas spatiotemporal clustering-based forecasting is more suitable

Conclusions

In this study, based on the spatiotemporal clustering analysis of the AQI in Beijing from 2015 to 2016, the CNN, the LSTM, and the CNN-LSTM were developed as multiple-hour and multiple-site forecasting models. The performance of these models was compared with the BPNN. The major findings are as follows:

- In the temporal dimension, there were obvious seasonal differences of the AQI in the two years, decreasing in the order of winter, spring, autumn, and summer, showing a U-shaped variation on ...

...

CRedit authorship contribution statement

Rui Yan: Conceptualization, Methodology, Software, Writing - original draft, Project administration. **Jiaqiang Liao:** Methodology, Data curation. **Jie Yang:** Methodology, Software. **Wei Sun:** Writing - review & editing, Funding acquisition, Supervision. **Mingyue Nong:** Investigation, Resources. **Feipeng Li:** Investigation, Resources. ...

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

Acknowledgements

This study is financially supported by Top-Notch Young Talents of Pearl River Talents Plan (2019QN01G106), the Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai) (99147-42080011), the Hundred Talents Program of Sun Yat-Sen University (37000-18841201), National Undergraduate Training Programs for Innovation and Entrepreneurship (201901211), and the National Program on Key Research Projects of China (2017YFC1502706). ...

References (66)

N.i. Bai *et al.*

[The pharmacology of particulate matter air pollution-induced cardiovascular dysfunction](#)

[Article preview](#)[Cited by \(442\)](#)[Metrics](#)

[propagation neural network based on wavelet decomposition with meteorological conditions](#)

Atmospheric Pollution Research (2016)

A.K. Bhunia *et al.*

[Script identification in natural scene image and video frames using an attention based Convolutional-LSTM network](#)

Pattern Recognition (2019)

M. Boznar *et al.*

[A neural network-based method for short-term predictions of ambient SO₂ concentrations in highly polluted industrial areas of complex terrain](#)

Atmospheric Environment. Part B. Urban Atmosphere (1993)

B. Brunekreef *et al.*

[Air pollution and health](#)

The Lancet (2002)

M. Cai *et al.*

[Prediction of hourly air pollutant concentrations near urban arterials using artificial neural network approach](#)

Transportation Research Part D: Transport and Environment (2009)

S. Chen *et al.*

[A hybrid PSO-SVM model based on clustering algorithm for short-term atmospheric pollutant concentration forecasting](#)

Technological Forecasting and Social Change (2019)

W. Chen *et al.*

[Diurnal, weekly and monthly spatial variations of air pollutants and air quality of Beijing](#)

Atmospheric Environment (2015)

J. Cui *et al.*

[A framework for investigating the air quality variation characteristics based on the monitoring data: Case study for Beijing during 2013–2016](#)

Journal of Environmental Sciences (2019)

H. Esen *et al.*

[Artificial neural networks and adaptive neuro-fuzzy assessments for ground-coupled heat pump system](#)

Energy and Buildings (2008)

Multi-hour and multi-site air quality index forecasting in Beijing using CNN, LSTM, CNN-LSTM, and spatiotemporal...

Article preview

Cited by (442)

Metrics

☆ This article has been certified as Reproducible by the CodeOcean:
<https://codeocean.com/> ↗

[View full text](#)

Recommended articles

No articles found.

Cited by (442)

Review article

Deep learning for air pollutant concentration prediction: A review

Zhang B., ..., Pan J.

Atmospheric Environment • Volume 290 • 2022 • Article 119347

[Show abstract](#) ✓

Comparative Analysis of Multiple Deep Learning Models for Forecasting Monthly Ambient PM2.5 Concentrations: A Case Study in Dezhou City, China ↗

He Z., Guo Q.

Atmosphere • Volume 15 • 2024 • Article 1432

Machine learning algorithms to forecast air quality: a survey ↗

Méndez M., ..., Nunez M.A.

Artificial Intelligence Review • Volume 56 • 2023

Prediction of Monthly PM2.5 Concentration in Liaocheng in China Employing Artificial Neural Network ↗

He Z., ..., Li X.

Atmosphere • Volume 13 • 2022 • Article 1221

A systematic literature review of deep learning neural network for time series air quality forecasting ↗

Zaini N.B., ..., Abdul Malek M.

Environmental Science and Pollution Research • Volume 29 • 2022

Short-Term Solar Power Predicting Model Based on Multi-Step CNN Stacked LSTM Technique ↗

Elizabeth Michael N., ..., Al-Durra A.

Energies • Volume 15 • 2022 • Article 2150



[View all citing articles on Scopus](#) ↗

Metrics

Multi-hour and multi-site air quality index forecasting in Beijing using CNN, LSTM, CNN-LSTM, and spatiotempor...

Article preview	Cited by (442)	Metrics
Captures		
Mendeley Readers		310
		View details ↗

© 2020 Elsevier Ltd. All rights reserved.



All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.

